The Tool Switching Problem and its Neighbours: a mid-internship report

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Abstract

This internship was set up to study the Job Sequencing and Tool Switching Problem (SSP) and its ties to other combinatorial-optimization problems, without a fixed objective. This report gives a mid-point account of four related lines of work that grew out of that remit: a study of how far the standard group-then-sequence heuristic can be from optimal; the search for a formulation of the SSP that a branch-and-price can actually handle; a Benders decomposition that exploits the problem's polynomial inner subproblem; and a recent, still-open polyhedral view of the grouping structure. The threads are at different stages: some carry proofs, some carry computational evidence I am not yet ready to call conclusive, and one is barely started. I have tried to be exact about which is which.

1 Starting point and the shape of the work

This internship did not begin with a problem to solve. The remit was to study the Job Sequencing and Tool Switching Problem (SSP) and how it relates to other problems in combinatorial optimization—a licence to explore rather than a target to reach. What follows is a report on where that exploration has gone so far.

The entry point was a thesis defence. In my first days I attended Felipe Otiai's defence of a branch-and-price algorithm for the *Job Grouping Problem* (JGP)—the problem of packing a set of jobs into as few tool configurations as possible [9]. What makes his method work is the strength of its set-covering relaxation: the linear bound is so close to the integer optimum that the search certifies optimality in a handful of nodes, and it comfortably beats the compact formulations it is measured against. The question I was handed was the obvious one: can the same be done for the SSP?

That question is deceptively simple, because the SSP is the more demanding relative of the problem Felipe solved. The JGP only asks *which* jobs can share a magazine; the SSP also asks in *what order* to run them, so as to minimise the number of tools swapped in and out along the way. As I explain in Section 2, for a fixed order the swapping is easy—an old and exact greedy rule settles it—so all of the difficulty is in the ordering, and it is genuinely hard. Trying to carry a branch-and-price across from the JGP therefore turned out to be neither routine nor a dead end, but a way into several different questions. Those questions, posed over the first months by my two advisors, are the four threads of this report.

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Thread A — how much does grouping-then-sequencing lose? The most natural heuristic for the SSP first solves the JGP to group the jobs, then orders the groups. Both advisors wanted to know how safe that decomposition is: how far from optimal can it be, when is it exact, and can one choose the grouping cleverly to do better? A subtlety shaped the whole thread. The standard grouping formulation returns a *single* optimal grouping, but the questions are really about the *set* of optimal groupings—which one to sequence, and whether a non-minimal grouping can beat the minimal ones—so I was asked to enumerate them. I did this first with the polytope-vertex enumerator PORTA and then, once PORTA stopped scaling past about eight jobs, with a small breadth-first enumerator written for the purpose. Section 3 reports what that pool of optima reveals: an exactness theorem, a smallest counterexample, an unbounded-gap construction, a worst-case ratio, and two conjectures I have tested but not proved.

Thread B — a formulation one can actually solve. Running the other way, toward exact solution, the first obstacle was the model itself. The configuration-based formulation I was given [3] is a generalised travelling-salesman model whose subtour elimination, as originally written, is not correct; the fix (via prize-collecting-TSP constraints, worked out in Moreira [7]) restores correctness but leaves the model exponential in *both* its constraints and its variables—the constraints being the concern for a branch-and-price. I was asked to look for a formulation with only polynomially many constraints, in the number of jobs or tools rather than configurations. An MTZ-style attempt got part of the way but could not shed the exponential rows; that failure is what led to the position-indexed formulations of Section **??**, which do keep a fixed, polynomial row set and over which column generation runs. This thread is the most recent and the least settled: the formulations and their pricing are worked out and numerically checked, but I have not yet taken them back to Dr. Colares.

Thread C — Benders, because the inner problem is easy. Dr. Colares also suggested a decomposition in the opposite spirit to column generation. Since the hard part of the SSP is the ordering and the loading is polynomial, a Benders scheme can put the order in a master problem and settle the loading exactly in the subproblem, generating cuts from it. The idea followed a successful use of Benders on a vehicle-routing problem in the group; here the inner problem is even more convenient, being the exact greedy rule of Section 2. Section **??** describes the resulting branch-and-Benders-cut solver.

Thread D — a polyhedral view, just opened. Most recently, while I was walking Prof. Wagler through the six-job example that opens Thread A, the discussion turned to the combinatorial structure behind it—the clutter and blocker of the configurations that can serve each job, and the hypergraph they form. This is a mathematician's reading of the same objects, and it is only one conversation old; Section **??** records it as an open direction, with one observation from Thread A (about when the underlying graph is perfect) as a possible starting point.

The threads are not equally mature, and I have not forced them into a single story they do not yet have. Thread A is the most complete; Thread C is implemented and verified; Thread B is promising but unvalidated; Thread D is a hunch. Where I state a theorem it is proved; where I rely on computation I say what was tested; where I am uncertain I say so. Section **??** closes with what each thread needs next.

2 Background

This section fixes the problem, the greedy loading rule, the grouping view, and the one geometric picture—jobs as clusters of configurations—that all four threads share. A reader who knows the SSP can skim to Section 3.

2.1 The problem

A single machine must process n jobs $J = \{1, \dots, n\}$. Processing job j requires a set of tools $T_j \subseteq T$, drawn from a library of $m = |T|$ tools, to be mounted together in the machine's magazine. The magazine holds only $b < m$ tools at once and $|T_j| \le b$, so as the jobs run in some order the magazine must be reloaded, one tool swapped for another. The SSP asks for the order, and the accompanying sequence of magazine contents, that minimises the total number of tools loaded.

Definition 1 (Schedule and cost)**.** A *schedule* is a permutation π of the jobs together with magazine contents $M_1, \dots, M_n$ with $|M_i| \le b$ and $T_{\pi(i)} \subseteq M_i$. Starting from an empty magazine ($M_0 = \varnothing$), its cost is the number of insertions, $\sum_{i=1}^{n} |M_i \setminus M_{i-1}|$; the minimum over all schedules is Z^*.

Example 1 (The six-job ring). One small instance runs through the report. In the 6-*ring*, six jobs use six tools with $b = 3$ and $T_j = \{j, j+1\}$ (indices modulo 6), so consecutive jobs share one tool and non-consecutive jobs share none (Figure 1). Running the jobs in the cyclic order $1, \dots, 6$ and loading by the rule of the next subsection, the magazine passes through $\{1,2,3\}, \{1,2,3\}, \{1,3,4\}, \{1,4,5\}, \{1,5,6\}, \{1,5,6\}$: three insertions to build the first magazine, then one at each of positions $3, 4, 5$, for a cost of 6 (or 3 in the convention of Section 2.5 that does not charge the first load). This is optimal, yet the two-phase heuristic returns 7 (Section 3), and the 6-ring is the smallest instance on which it is beaten at all.

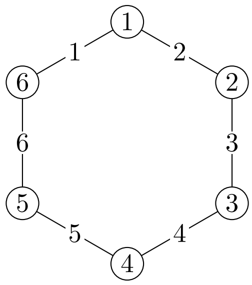

Figure 1: The 6-ring: six jobs on a cycle, each edge labelled by the single tool its two endpoints share.

2.2 Loading is easy; ordering is hard

Fix the order. Then the best way to fill the magazine is known exactly and cheaply.

Proposition 1 (12)**.** *For a fixed order, the loading of minimum cost is obtained by the* Keep Tool Needed Soonest *rule: whenever a tool must be brought in and the magazine is full, remove the one whose next use lies furthest ahead. It runs in* $O(n|T|)$.

This is worth pausing on, because it is the pivot of the whole internship. The loading subproblem is exactly the offline caching problem—the magazine is a cache of size b, each job is a request for a set of items, and a reload is a cache miss—and the rule above is Bélády's optimal replacement policy [1]. Caching with a *fixed* request order is polynomial; the SSP is hard only because it is allowed to *reorder* the requests. That single fact—an easy inner problem inside a hard ordering problem—is what makes both a Benders decomposition (put the order outside, solve the loading inside; Thread C) and a column-generation model (Thread B) natural, and it is why $Z^* = \min_\pi \mathrm{KTNS}(\pi)$ is a sequencing problem rather than a loading one.

2.3 Grouping, and the heuristic

Ignore order for a moment and ask only which jobs can share a magazine.

Definition 2 (Grouping problem)**.** A *group* is a set of jobs whose tools jointly fit, $|\bigcup_{j\in G} T_j| \le b$. The Job Grouping Problem partitions the jobs into the fewest groups, K^*; it is the problem Felipe's branch-and-price solves [4, 6, 9].

Grouping suggests an obvious two-phase heuristic for the SSP: group the jobs (solve the JGP), then decide the order in which to run the groups (a small travelling-salesman problem over the group configurations), then read off a schedule. Call its cost H. Since it produces a valid schedule, $H \ge Z^*$; how much larger it can be is Thread A.

2.4 One picture: jobs as clusters of configurations

The two problems, and every formulation below, live most naturally in the space of *configurations.*

Definition 3. A *configuration* is a full magazine, a set $C \subseteq T$ with $|C| = b$. For a job j, the configurations that can serve it form its *cluster* $H_j = \{C : T_j \subseteq C\}$. Moving from C to C' costs $d(C, C') = |C' \setminus C|$ insertions.

In this picture a schedule is a walk through configurations that visits at least one member of each cluster H_j, and its cost is the total of the step costs d. The SSP is thus a travelling-salesman problem through overlapping clusters—a generalised TSP, except that the clusters overlap heavily (a rich configuration serves many jobs), which is exactly what makes it richer than a textbook GTSP [5, 8]. This is the formulation Dr. Colares proposed as a starting point [3], and it is where Thread B begins: as it stands the model needs one subtour-elimination constraint per subset of configurations, and even after the correction that makes it valid [7] the count is exponential—the difficulty Section **??** sets out to remove.

2.5 Two lower bounds, used everywhere

Two easy bounds recur through every thread, so I record them once. Both are stated in the convention that does not charge the first, unavoidable magazine load; the two conventions differ by the constant $\min(b, |U|)$, where $U = \bigcup_j T_j$ is the set of tools that are used at all.

Proposition 2. *$Z^* \ge K^* - 1$ and $Z^* \ge |U| - b$.*

Proof. Any schedule's magazine, with equal consecutive contents merged, is a walk through configurations covering every job; such a covering needs at least K^* distinct configurations, hence at least $K^* - 1$ steps, each costing ≥ 1. Separately, every tool in U must enter the magazine at some point, and only b can be present at the start, so at least $|U| - b$ insertions occur. □

The second bound, $|U| - b$, deserves a flag now because it becomes a theme: it is the value every configuration relaxation in Thread B collapses to, and Thread A shows it is often the exact optimum—the same coincidence, seen from two sides, that Section **??** argues is the crux of the whole problem.

3 Thread A: how far can grouping-then-sequencing be from optimal?

Both advisors asked the same question from different angles: the two-phase heuristic is the natural attack on the SSP, so what does it give up by fixing the grouping before the order? The

6-ring already shows it gives up something (Example 1: cost 7 against the optimum 6). This thread pins down what. I identify the exact source of the loss, show it can be made arbitrarily large, bound it in the worst case, and state two sharper limits supported by computation but not yet proved. What is called a theorem here is proved; the two conjectures rest only on enumeration, and I say so where they appear.

3.1 Method: the pool of optimal groupings

The heuristic solves the JGP for one minimum grouping, but every question here concerns the *set* of optimal groupings—which minimum grouping to sequence, and whether a non-minimum grouping can do better. The formulation I was given returns a single optimum, so I enumerate the rest directly, treating the integral feasible groupings as objects to list: first with PORTA, which enumerates the vertices of a polyhedron exactly, and—once PORTA stopped scaling past about eight jobs—with a breadth-first enumerator written for the task. The pool of optimal and near-optimal groupings this produces is the raw material for the rest of the thread, and, as Section **??** notes, it is also where Prof. Wagler's polyhedral reading of the problem begins.

3.2 The loss is the price of grouping minimally

Attach to a grouping the cost of sequencing it optimally: for a partition $\mathcal{P}$ into groups with chosen configurations $C_1, \ldots, C_p$, let $\gamma(\mathcal{P}) = \min_\sigma \sum_l d(C_{\sigma(l)}, C_{\sigma(l+1)})$ be the cheapest walk over orderings σ. The heuristic evaluates γ on a *minimum-cardinality* grouping. Dropping that restriction makes the configuration picture exact.

Theorem 1 (grouping exactness). *$Z^* = \min_{\mathcal{P}} \gamma(\mathcal{P})$, the minimum over* all *feasible groupings.*

Proof. Any grouping, run group by group in its cheapest order, is a schedule of cost $\gamma(\mathcal{P})$, so $Z^* \leq \min_{\mathcal{P}} \gamma(\mathcal{P})$. Conversely, take an optimal schedule and merge equal consecutive magazines into a trajectory $D_1, \ldots, D_r$; the jobs sharing each D_l form a group, so this is a feasible grouping whose ordering $D_1, \ldots, D_r$ costs exactly Z^*, giving $\min_{\mathcal{P}} \gamma(\mathcal{P}) \leq Z^*$. □

Corollary 1. *With $H = \min\{\gamma(\mathcal{P}) : |\mathcal{P}| = K^*\}$ the best two-phase cost, $H - Z^* = \min_{|\mathcal{P}|=K^*} \gamma - \min_{\mathcal{P}} \gamma$; the gap is positive exactly when no minimum-cardinality grouping is a cheapest grouping.*

The identity was confirmed on 1,306 instances (the $b = 3$ two-tool family and 800 random mixed instances) with no exception. It is exactly what happens on the ring: the optimum uses *four* configurations, $\{1, 2, 3\} \to \{1, 3, 4\} \to \{1, 4, 5\} \to \{1, 5, 6\}$, each step changing one tool, for cost 3; every minimum grouping uses only $K^* = 3$ configurations (say $\{1, 2, 3\}, \{3, 4, 5\}, \{5, 6, 1\}$, any two differing in two tools) and costs 4. The loss of one is the corollary's difference—the cheaper grouping is not minimum, so the grouping phase never offers it.

3.3 The loss is unbounded, but bounded in ratio

That the gap can be nonzero is one thing; that it can be arbitrarily large is another. Table 1 lists the k-ring for $k = 3, \ldots, 8$: the gap first appears at $k = 6$. Repeating the worst ring drives the gap up without limit.

Proposition 3 (unbounded gap). *For g tool-disjoint copies of the 6-ring, $Z^* = 6g - 3$ and $H = 7g - 3$, so the gap is g; the ratio $(7g - 3)/(6g - 3)$ falls from $4/3$ at $g = 1$ towards $7/6$.*

Proof. The copies are disjoint, so $|U| = 6g$ and $Z^* \geq 6g - 3$ by Proposition 2; processing the copies one after another attains $6g - 3$. The heuristic uses $3g$ configurations, three per copy at pairwise distance 2 (so 4 within a copy) and one disjoint transition of cost 3 between consecutive copies, giving $4g + 3(g - 1) = 7g - 3$. Verified for $g = 1, \ldots, 4$. □

k	K^*	Z^*	H	gap	H/Z^*
3	1	0	0	0	—
4	2	1	1	0	1
5	3	2	2	0	1
6	3	3	4	1	4/3
7	4	4	5	1	5/4
8	4	5	6	1	6/5

Table 1: The k-ring ($b = 3$, free-initial costs).

In ratio, however, the heuristic is never worse than a constant factor.

Proposition 4 (b-approximation)**.** *$H \leq b(K^* - 1)$, so $H/Z^* \leq b$.*

Proof. A minimum grouping has K^* configurations of size b, so any ordering makes $K^* - 1$ transitions of at most b tools each; divide by $Z^* \geq K^* - 1$ (Proposition 2). □

3.4 Two conjectures

The true worst case looks much smaller than b, but I have not been able to prove it. The enumeration of Section 3.1 supports two statements.

Conjecture 1. For $b = 3$, $H/Z^* \leq 4/3$.

Tested by exhaustive enumeration of the $b = 3$ two-tool family with $m \leq 6$ (10,691 instances) and of mixed size-$\{2, 3\}$ jobs with $m \leq 5$ (6,065 more): the maximum ratio is exactly 4/3, reached only by the 6-ring, with no counterexample.

Conjecture 2. For $K^* \geq 2$, the gap is at most $K^* - 2$.

No violation appeared over about 1,380 enumerated instances; it is consistent with the zero gap for $K^* \leq 2$ and with Proposition 3, where $K^* = 3g$ and $g \leq 3g - 2$. I record both honestly as open: the empirical regularity is strong, but these are exactly the kind of ratio statements that interact with the cost convention, so I would not present them as more than conjectures. One caution is already clear—the constant 4/3 is specific to $b = 3$; the extremal ratio does not decrease with b, and reaches 3/2 at $b = 5$.

3.5 When the gap is zero

Two structural remarks come out of the same analysis. First, a certificate: since $H \geq Z^* \geq \max(K^* - 1, |U| - b)$, any instance on which the heuristic meets a lower bound has zero gap, which already settles the k-rings for $k \leq 5$. Second, a warning against a natural but false hope—that polyhedral regularity controls the gap. Form the graph on the jobs joining two that cannot share a configuration; for the 6-ring it is the triangular prism, a perfect graph, and the gap is one, while for the 5-ring it is the odd cycle C_5, imperfect [2], and the gap is zero. Perfectness and the gap are independent. (This graph is one of the objects Prof. Wagler proposed to study; Section **??**.)

3.6 A variant that collapses to grouping

The advisors also asked whether some cost variant makes the two-phase heuristic exact. It does, under a setup cost. Charge, besides the tool switches, a fixed $\rho \geq 0$ per change of configuration—the ratio of magazine-setup to tool-switch time. A grouping with p configurations makes $p - 1$ changes, so as ρ grows the setup term forces the optimum onto minimum-cardinality groupings, the heuristic's own domain.

Proposition 5 (collapse)**.** *If $\rho > H - Z^*$, every optimum of the setup-augmented problem uses a minimum-cardinality grouping, so the heuristic is exact; the threshold satisfies $\rho_c \leq H - Z^*$, and $\rho_c = 1$ for the rings.*

Proof. A grouping $\mathcal{P}$ has $|\mathcal{P}| \geq K^*$ and (Theorem 1) $\gamma(\mathcal{P}) \geq Z^*$, so its augmented cost $\rho(|\mathcal{P}| - 1) + \gamma(\mathcal{P})$ exceeds the best minimum grouping's $\rho(K^* - 1) + H$ whenever $\rho(|\mathcal{P}| - K^*) > H - Z^*$, which holds for all $|\mathcal{P}| > K^*$ once $\rho > H - Z^*$. □

This is not idle: the setup-to-switch ratios reported for real tooling are of order 3–60 [10, 11], well above the ring threshold, so on ring-like instances the heuristic is already exact for the setup objective.

3.7 Which grouping to run—and what I am unsure of

Theorem 1 turns the whole thread into a concrete algorithmic question. The optimum is realised by *some* grouping, and minimum ones can miss it, so one should ask which grouping to sequence—on the ring, admitting a single extra configuration already recovers the optimum. Characterising the cheapest groupings, or finding a rule that improves on the minimum-cardinality choice without solving the SSP outright, is open, and it is the most direct route to a heuristic with a guarantee. I will be candid that this thread is not yet where I want it: the exactness theorem and the unbounded-gap construction are solid, but the two conjectures and the grouping-selection rule are not settled, and turning the empirical 4/3 into a theorem (or a counterexample) is what would make the thread publishable.

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